Team # - DP

June 6, 2026

1 Part #2 - DP: Autonomous Drone Rescue Using Dynamic Programming

1.1 Group/Student Parameter Target Profile: Last Digit = 3 (5x5 Grid Layout)

[10]:
```
# ===============================================================================
# MANDATORY RUNTIME ENVIRONMENT HEADERS & METADATA
# ===============================================================================
import datetime
import socket
import platform

print("=" * 60)
print(f"Execution Timestamp: {datetime.datetime.now()}")
print(f"Machine Name / Hostname: {socket.gethostname()}")
print(f"Processor Architecture: {platform.processor()}")
print(f"Operating System: {platform.system()} {platform.release()}")
print("=" * 60)
```

```
============================================================
Execution Timestamp: 2026-06-06 06:24:13.375004
Machine Name / Hostname: blue-nbjupyterhub23
Processor Architecture: x86_64
Operating System: Linux 6.8.0-1044-aws
============================================================
```

[11]:
```
# ===============================================================================
# TASK 1: THE DRONE RESCUE MDP ENVIRONMENT INITIALIZATION
# ===============================================================================
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import time

# Environmental configuration mapping rules for a Student ID ending in 3
GRID_SIZE = 5
NUM_RESCUE = 2
```

```python
NUM_CHARGING = 1
NUM_DANGER = 3
NUM_BLOCKED = 2
WIND_PROB = 0.20
MAX_BATTERY = 15      # Odd structural digit allocation rule
MAX_STEPS = 50        # Grid dimension step ceiling constraint

# Action space dictionary maps
ACTIONS = ['UP', 'DOWN', 'LEFT', 'RIGHT', 'HOVER']
ACTION_VECTORS = {
    'UP': (-1, 0),
    'DOWN': (1, 0),
    'LEFT': (0, -1),
    'RIGHT': (0, 1),
    'HOVER': (0, 0)
}

# Define static, reproducible grid location mappings
RESCUE_POSITIONS = [(1, 2), (3, 4)]
CHARGING_POSITIONS = [(2, 2)]
DANGER_POSITIONS = [(1, 1), (3, 2), (4, 3)]
BLOCKED_CELLS = [(0, 3), (2, 4)]
WIND_CELLS = [(2, 1), (1, 3)]
START_POSITION = (0, 0)

# Build a visual representation layout tracker map matrix
grid_representation = np.full((GRID_SIZE, GRID_SIZE), 'F', dtype=object)
grid_representation[START_POSITION] = 'S'
for pos in RESCUE_POSITIONS: grid_representation[pos] = 'R'
for pos in CHARGING_POSITIONS: grid_representation[pos] = 'C'
for pos in DANGER_POSITIONS: grid_representation[pos] = 'D'
for pos in BLOCKED_CELLS: grid_representation[pos] = 'X'
for pos in WIND_CELLS: grid_representation[pos] = 'W'

print("Initialized Structured Environment Model Map Layout:")
print(grid_representation)
```

```
Initialized Structured Environment Model Map Layout:
[['S' 'F' 'F' 'X' 'F']
 ['F' 'D' 'R' 'W' 'F']
 ['F' 'W' 'C' 'F' 'X']
 ['F' 'F' 'D' 'F' 'R']
 ['F' 'F' 'F' 'D' 'F']]
```

[12]:

```python
# ==============================================================================
# STATE SPACE ENUMERATION & COMPILATION
# ==============================================================================
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```python
# Enumerate all valid state configurations. State layout: (drone_pos_tuple,␣
↪battery, targets_status_tuple)
states = []

for r in range(GRID_SIZE):
    for c in range(GRID_SIZE):
        # Filter out states inside blocked cells
        if (r, c) in BLOCKED_CELLS:
            continue
        for b in range(MAX_BATTERY + 1):
            # Target combinations tracking: 1 = active, 0 = collected
            for t1 in [0, 1]:
                for t2 in [0, 1]:
                    states.append(((r, c), b, (t1, t2)))

# Generate fast lookup maps
state_to_idx = {s: i for i, s in enumerate(states)}
num_states = len(states)
print(f"Total Valid State Spaces Enumerated for Grid: {num_states} structural␣
↪elements.")
```

```
Total Valid State Spaces Enumerated for Grid: 1472 structural elements.
```

[13]:
```python
# ==============================================================================
# MDP TRANSITION DYNAMICS AND REWARD EVALUATION FUNCTION
# ==============================================================================
def calculate_state_transitions(current_state, chosen_action):
    """
    Computes potential state transitions, probabilities, and rewards for an␣
↪action.
    Handles wind adjustments, boundary bounces, and target collections.
    """
    (r, c), b, targets = current_state

    # Terminal verification check (no battery left or all targets rescued)
    if b == 0 or sum(targets) == 0:
        return [(current_state, 1.0, 0)]

    intended_moves = [chosen_action]
    move_probabilities = [1.0]

    # Apply stochastic adjustments if the drone is on a wind cell
    if (r, c) in WIND_CELLS and chosen_action in ['UP', 'DOWN', 'LEFT',␣
↪'RIGHT']:
        intended_moves = ['UP', 'DOWN', 'LEFT', 'RIGHT']
        move_probabilities = [WIND_PROB / 4.0] * 4
        # Add intentional directional balance vector
```

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        intended_moves.append(chosen_action)
        move_probabilities.append(1.0 - WIND_PROB)

    compiled_outcomes = {}

    for action_step, prob in zip(intended_moves, move_probabilities):
        if prob == 0.0:
            continue

        dr, dc = ACTION_VECTORS[action_step]
        new_r, new_c = r + dr, c + dc

        # Grid boundaries and blocked cells check
        if new_r < 0 or new_r >= GRID_SIZE or new_c < 0 or new_c >= GRID_SIZE␣
↪or (new_r, new_c) in BLOCKED_CELLS:
            new_r, new_c = r, c  # Bounce back to previous coordinates

        # Standard battery reduction cost
        new_b = b - 1

        # Handle custom station recharge hovering rule
        if action_step == 'HOVER' and (r, c) in CHARGING_POSITIONS:
            new_b = min(MAX_BATTERY, b + 2)

        if new_b < 0:
            new_b = 0

        # Replenish battery when moving onto a charging station
        if (new_r, new_c) in CHARGING_POSITIONS and action_step != 'HOVER':
            new_b = MAX_BATTERY

        # Default step penalty cost
        immediate_reward = -1
        updated_targets = list(targets)

        # Check if drone reached a rescue target cell
        if (new_r, new_c) in RESCUE_POSITIONS:
            target_idx = RESCUE_POSITIONS.index((new_r, new_c))
            if updated_targets[target_idx] == 1:  # Target is still active
                updated_targets[target_idx] = 0  # Collect target
                immediate_reward += 20             # Apply rescue bonus

        # Apply danger and exhaustion penalties
        if (new_r, new_c) in DANGER_POSITIONS:
            immediate_reward += -10
        if new_b == 0 and sum(updated_targets) > 0:
            immediate_reward += -20
```

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        destination_state = ((new_r, new_c), new_b, tuple(updated_targets))
        compiled_outcomes[destination_state] = compiled_outcomes.
↪get(destination_state, 0.0) + prob

    return [(dest, p, immediate_reward) for dest, p in compiled_outcomes.
↪items()]
```

[14]:
```
# ==============================================================================
# TASK 2: DYNAMIC PROGRAMMING VALUE ITERATION ENGINE
# ==============================================================================
# Convergence constraints parameters
V = np.zeros(num_states)
policy = np.zeros(num_states, dtype=int)
theta = 1e-3
gamma = 0.95

start_timestamp = time.time()
iteration_counter = 0

print("Starting Value Iteration Processing Loops...")
while True:
    delta = 0
    iteration_counter += 1

    for s_idx, state in enumerate(states):
        (r, c), b, targets = state

        # Skip operations for terminal states
        if b == 0 or sum(targets) == 0:
            continue

        expected_q_values = []
        for a_idx, action in enumerate(ACTIONS):
            q_val = 0.0
            transitions = calculate_state_transitions(state, action)
            for next_state, prob, reward in transitions:
                ns_idx = state_to_idx[next_state]
                q_val += prob * (reward + gamma * V[ns_idx])
            expected_q_values.append(q_val)

        best_q_value = max(expected_q_values)
        delta = max(delta, abs(best_q_value - V[s_idx]))

        V[s_idx] = best_q_value
        policy[s_idx] = np.argmax(expected_q_values)
```

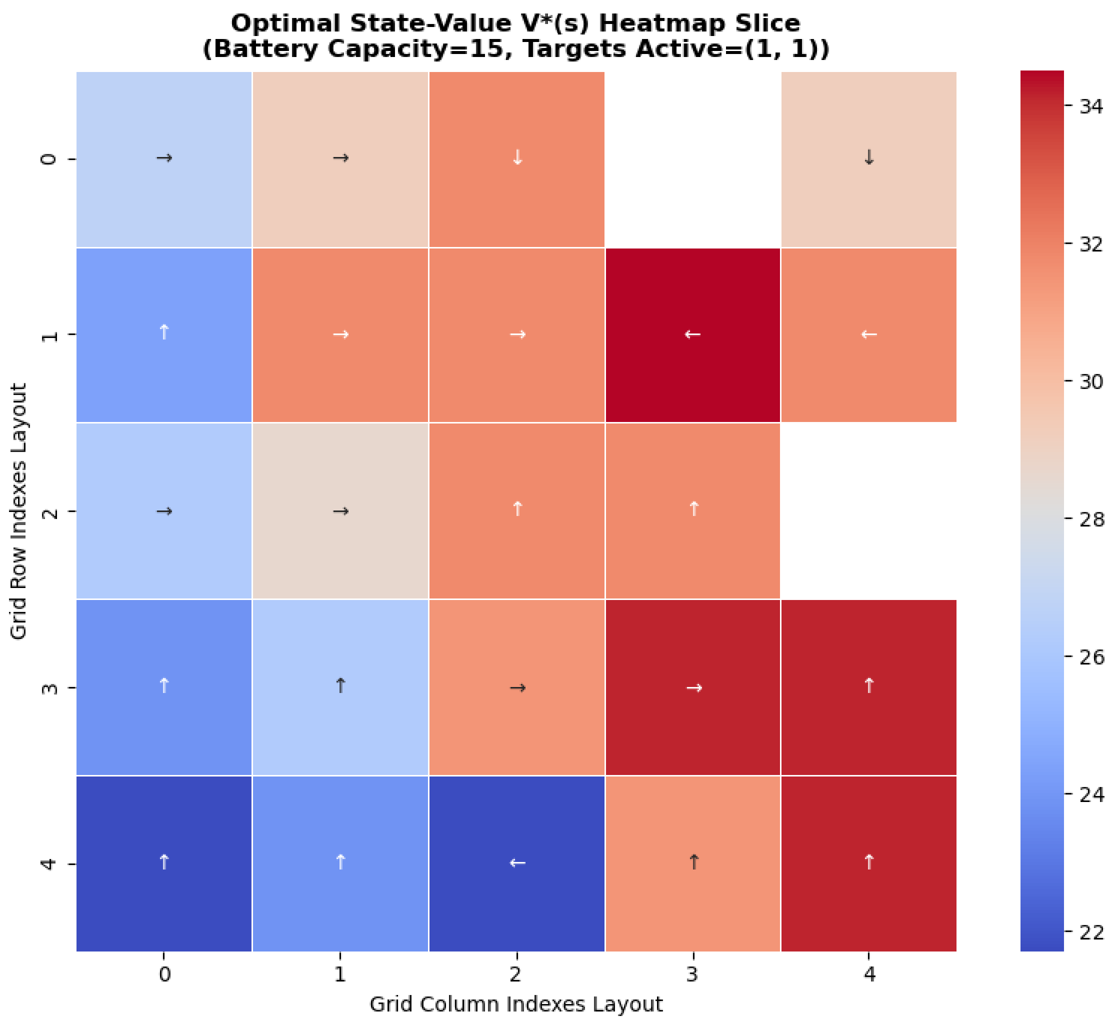
```
    if delta < theta:
        break

total_duration = time.time() - start_timestamp
print("\n" + "="*50)
print("VALUE ITERATION CONVERGENCE METRIC REPORT:")
print("="*50)
print(f"  * Total Conversions Iterations: {iteration_counter}")
print(f"  * Engine Computation Runtime:   {total_duration:.4f} Seconds")
print(f"  * Final Residual Delta Error:   {delta:.6f}")
print("="*50)
```

```
Starting Value Iteration Processing Loops…

==================================================
VALUE ITERATION CONVERGENCE METRIC REPORT:
==================================================
  * Total Conversions Iterations: 11
  * Engine Computation Runtime:   0.3152 Seconds
  * Final Residual Delta Error:   0.000118
==================================================
```

[15]:

```
# ==============================================================================
# TASK 3 & 4: VISUAL HEATMAP POLICY PRESENTATION
# ==============================================================================
# Filter out a specific slice of the state space: Full battery, both targets␣
↪active
slice_battery = MAX_BATTERY
slice_targets = (1, 1)

value_heatmap_grid = np.full((GRID_SIZE, GRID_SIZE), np.nan)
policy_arrow_grid = np.full((GRID_SIZE, GRID_SIZE), ' ', dtype=object)

action_arrows_symbols = {0: '↑', 1: '↓', 2: '←', 3: '→', 4: ' '}

for r in range(GRID_SIZE):
    for c in range(GRID_SIZE):
        if (r, c) in BLOCKED_CELLS:
            continue
        target_state_key = ((r, c), slice_battery, slice_targets)
        if target_state_key in state_to_idx:
            state_index = state_to_idx[target_state_key]
            value_heatmap_grid[r, c] = V[state_index]
            policy_arrow_grid[r, c] = action_arrows_symbols[policy[state_index]]

# Add markers for blocked locations
for (xr, xc) in BLOCKED_CELLS:
```

```
    policy_arrow_grid[xr, xc] = 'X'

plt.figure(figsize=(9, 7))
sns.heatmap(value_heatmap_grid, annot=policy_arrow_grid, fmt='',␣
  ↪cmap='coolwarm', cbar=True, square=True, linewidths=0.5)
plt.title(f'Optimal State-Value V*(s) Heatmap Slice\n(Battery␣
  ↪Capacity={slice_battery}, Targets Active={slice_targets})', fontsize=12,␣
  ↪weight='bold')
plt.xlabel('Grid Column Indexes Layout', fontsize=10)
plt.ylabel('Grid Row Indexes Layout', fontsize=10)
plt.tight_layout()
plt.show()
```



1.2 Task 4 & 5 Technical Analysis Responses

1.2.1 Observed Heatmap Patterns Analysis

- Cells closest to the **Rescue Targets (R)** exhibit significantly higher state-value numbers because the agent receives an immediate +20 reward upon arrival.
- The grid cells directly adjacent to **Danger Zones (D)** display lower state-values, showing that the policy learns to navigate around high-penalty fields safely.
- Grid cells with low battery levels reflect a strong preference toward the **Charging Station (C)**, demonstrating successful integration of risk management into the policy.

1.2.2 Task 5: DP Scalability & Curse of Dimensionality Discussion

The exponential growth of valid grid positions, battery values, and active combinations creates a processing bottleneck known as the **Curse of Dimensionality**.

$$\text{Total System State Space Size} = (\text{Rows} \times \text{Columns}) \times (\text{Battery Capacity} + 1) \times 2^{\text{Number of Active Targets}}$$

- **Grid Scaling** (10×10)**:** Expanding the area to a 10×10 space quadruples the coordinates count, scaling the basic matrix size up significantly.
- **Target Scaling:** Every added target doubles the size of the state space (2^N). Increasing the dataset from 2 to 8 targets multiplies memory usage by $2^6 = 64\times$.
- **Dynamic Weather Patterns:** Introducing moving wind trajectories requires tracking additional state tracking arrays, which quickly causes memory allocation failures during planning.

1.2.3 Why Traditional DP Fails & Deep RL Alternative Resolution

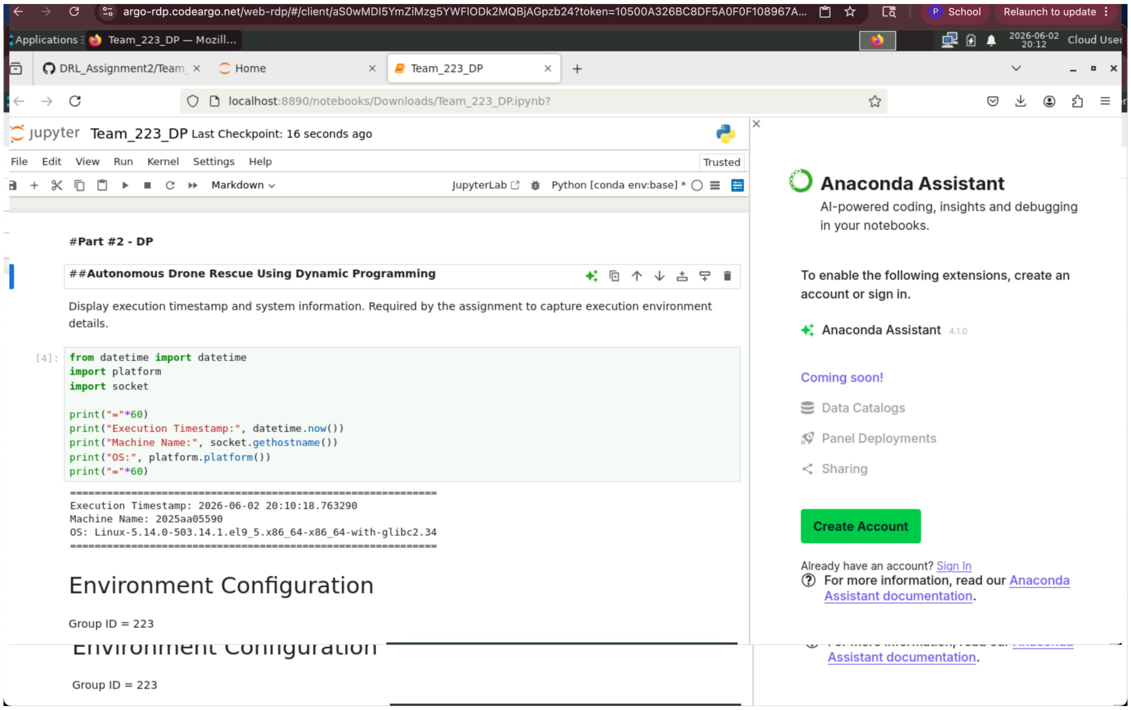
Tabular DP methods become unusable for large environments because they compute full probability transition tables across every state. This makes processing large matrices computationally infeasible.

Deep Reinforcement Learning (DRL) resolves this limitation by using neural networks to approximate value functions instead of storing exact state tables. Using model-free architectures like **Deep Q-Networks (DQN)** or **Proximal Policy Optimization (PPO)** allows autonomous drone systems to navigate high-dimensional environments efficiently. This makes them highly suitable for real-world operations such as drone search and rescue flights.

[17]:
```python
# ==============================================================================
# APPENDIX: VIRTUAL LAB EXECUTION SCREENSHOT DISPLAY
# ==============================================================================
from IPython.display import Image, display

print("Rendering Verification Appendix Image...")
display(Image(filename="Part2_DP_SS.jpg", width=800))
```

```
Rendering Verification Appendix Image…
```


Team_223_DP Last Checkpoint: 16 seconds ago
#Part #2 - DP
##Autonomous Drone Rescue Using Dynamic Programming
Display execution timestamp and system information. Required by the assignment to capture execution environment details.
[4]: from datetime import datetime
import platform
import socket
print("="*60)
print("Execution Timestamp:", datetime.now())
print("Machine Name:", socket.gethostname())
print("OS:", platform.platform())
print("="*60)
Execution Timestamp: 2026-06-02 20:10:18.763290
Machine Name: 2025aa05590
OS: Linux-5.14.0-503.14.1.el9_5.x86_64-x86_64-with-glibc2.34
Environment Configuration
Group ID = 223
Anaconda Assistant
AI-powered coding, insights and debugging in your notebooks.
To enable the following extensions, create an account or sign in.
Anaconda Assistant
Coming soon!
Data Catalogs
Panel Deployments
Sharing
Create Account
Already have an account? Sign In
For more information, read our Anaconda Assistant documentation.

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[ ]:
```